

Report R15 — the tower in the attractor plane, and whether the resets are idle inside it

Krylov sector analysis of quantum cellular automata

Tier 1 analysis

2026-08-05

Abstract

In the right-hand panel of R9’s F1 a column of V+reset rules stands at $a_{\text{att}} = 1$ with b_{att} running from 1.41 to 1.89: a bounded number of attractors, the largest of which still grows exponentially. There are 68 such rules of the 160 V+reset ones. Their growth bases are not spread out — 61 of the 68 sit on five *exact* algebraic constants, φ and $\sqrt{3}$ taking 21 each. The natural explanation is that the dissipative channels never act inside the largest attractor, so what grows there is really the underlying unitary. **That is true for exactly 12 of the 68, and false for the other 56.** For the twelve it is true in the strongest sense: the largest attractor *is* the largest sector of the coherent parent, state for state, and all twelve sit at φ . For the other fifty-six the resets do work at $\Theta(N)$ sites — the count of firing sites grows linearly with system size, so this is not a boundary effect — and the exponential growth survives dissipation rather than avoiding it. Nine rules reach φ *with* firing resets, so the constant does not determine the mechanism. obc0, $N = 10$ for the census.

1 The tower

A rule is in the tower if it is V+reset, its attractor *count* is flat ($a_{\text{att}} = 1$ to within 0.02) and its largest attractor grows exponentially ($b_{\text{att}} > 1.02$). That is 68 rules.

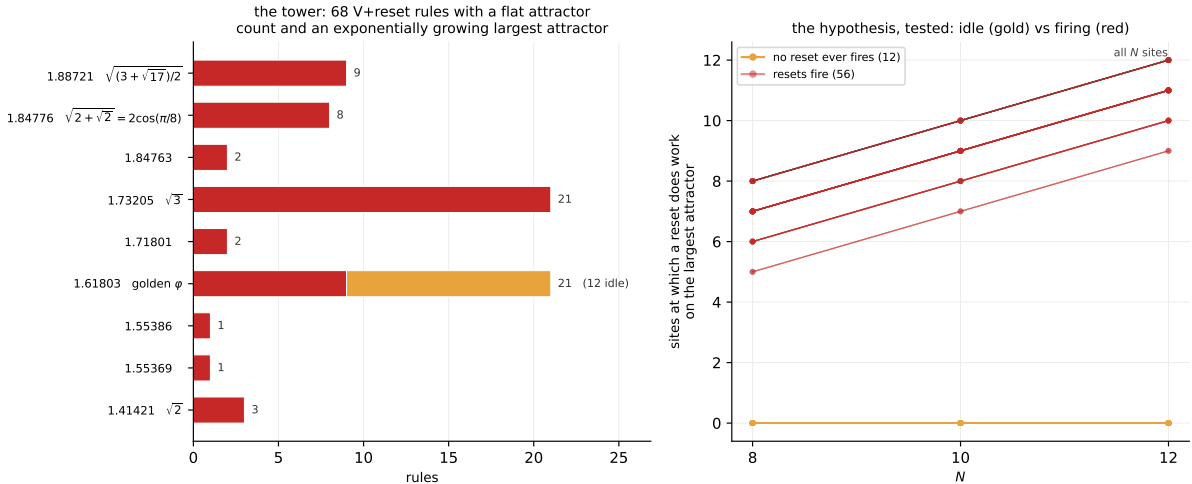


Figure 1: **Left:** the tower, one bar per distinct b_{att} , gold marking the rules on which no reset ever fires. **Right:** the hypothesis tested — the number of sites at which some reset does work on the largest attractor, against N . Flat at zero for twelve rules; linear in N for the rest, tracking the dotted “all N sites” line.

1.1 Where the bases sit

b_{att}	rules	exact	constant
1.414214	3	3	$\sqrt{2}$
1.553687	1	0	—
1.553861	1	0	—
1.618034	21	21	golden φ
1.718005	2	0	—
1.732051	21	20	$\sqrt{3}$
1.847632	2	0	—
1.847759	8	8	$\sqrt{2 + \sqrt{2}} = 2 \cos(\pi/8)$
1.887208	9	9	$\sqrt{(3 + \sqrt{17})/2}$

Five constants absorb 61 of the 68, and for those the base comes from an exact integer recurrence rather than a fit: $\sqrt{2}$ (3 rules), φ (21), $\sqrt{3}$ (20 of 21), $2 \cos(\pi/8) = \sqrt{2 + \sqrt{2}}$ (8) and $\sqrt{(3 + \sqrt{17})/2}$ (9). The remaining seven are fits. Two of them, at 1.847632, are within 1.3×10^{-4} of $2 \cos(\pi/8)$ and are almost certainly that constant imperfectly fitted; the pair at 1.5537/1.5539 and the pair at 1.7180 have no identification here and are left as fits rather than rounded into a name.

These are the constants of *constrained binary words* — Perron roots of small transfer matrices — which is exactly the company the unitary rules keep. That is what makes the idle-reset hypothesis so natural.

2 The hypothesis, tested

The dissipative channels never fire on the largest attractor, so what grows there is the underlying unitary.

A reset D writes 0 and E writes 1; either does *work* only when the target bit is not already at that value, and otherwise acts as the identity on that state. `tower.firing_sites` runs the brick-wall sweep by hand, following every Hadamard branch, and returns the sites at which some branch had a reset do work. (The hand sweep is validated against the engine’s successor map in the tests: it must reproduce `wcc.make_succ` exactly, over all states of 37 rules at three sizes, or the firing flags mean nothing.)

	rules	
no reset ever fires on the largest attractor	12	hypothesis holds
some reset fires	56	hypothesis fails

2.1 The twelve, and why they work

rule	tuple	b_{att}	$ A $	parent	parent D_{max}
1	VDDD	1.61803	144	W201	144
9	VIDD	1.61803	144	W201	144
65	VDID	1.61803	144	W201	144
110	IEIV	1.61803	55	W108	55
111	EEIV	1.61803	55	W108	55
124	IIEV	1.61803	55	W108	55
125	EIEV	1.61803	55	W108	55
126	IEEV	1.61803	55	W108	55
127	EEEV	1.61803	55	W108	55
129	VDDI	1.61803	144	W201	144
137	VIDI	1.61803	144	W201	144
193	VDII	1.61803	144	W201	144

For all twelve the statement holds in the strongest available form: the child’s successor map *coincides* with the coherent parent’s on every state of the largest attractor, and the attractor has exactly the size of the parent’s largest sector. It is not merely that the resets happen to be idle — the attractor *is* a sector of the parent, state for state.

They fall into two families of six, and the mechanism is visible in the tuple:

family	tuples	parent	$ A $ at $N = 10$
V at r_{00} , rest from $\{D, I\}$	VDDD, VIDD, VDID, VDDI, VIDI, VDII	W201	$144 = F_{12}$
V at r_{11} , rest from $\{E, I\}$	IEIV, EEIV, IIEV, EIEV, IEEV, EEEV	W108	$55 = F_{10}$

The reset’s target value agrees with what those sites already carry. In the first family the Hadamard fires only on the neighbourhood 00, and every D sits at a neighbourhood that occurs only when the site is already 0, so writing 0 changes nothing. The second family is the same statement with $0 \leftrightarrow 1$ and E for D. The surviving set is the parent’s Fibonacci-counted sector, hence φ — and indeed **every one of the twelve sits at φ** , none at any other constant.

2.2 The fifty-six, and why the hypothesis fails for them

For these the resets fire, and not marginally: the number of *sites* at which some reset does work on the largest attractor grows linearly with N (right panel of the figure), typically reaching $N - 1$ or N . So it cannot be rescued as a boundary effect or as a bounded set of defect sites — dissipation acts throughout the bulk of the attractor, and the exponential growth survives it rather than avoiding it.

What these rules share with the twelve is only the *form* of the answer: an algebraic Perron root. A reset that fires deterministically given its neighbourhood is still a local constraint, and a set of words cut out by local constraints is still counted by a transfer matrix. The constant does not require the dynamics to be unitary on the attractor — only local.

2.3 The constant does not identify the mechanism

Twenty-one rules sit at φ ; twelve are idle and **nine are not** — rules 11, 47, 81, 117, 139, 161, 171, 209 and 241 reach the same golden ratio with resets firing at $\Theta(N)$ sites. So one cannot read the mechanism off the growth base, which is worth saying because the temptation to do so is exactly what this report set out to test.

3 What this settles and what it does not

- **Settled.** The idle-reset explanation is a real mechanism and it is a minority one: 12 of 68, all at φ , all inheriting a parent sector outright.
- **Settled.** For the other 56 the resets act on $\Theta(N)$ sites of the attractor, so no version of “the channel is secretly unitary there” survives.
- **Open.** The transfer matrices themselves. $\sqrt{3}$, $2\cos(\pi/8)$ and $\sqrt{(3+\sqrt{17})/2}$ each take 8–21 rules, and the constraint each one counts has not been identified here; that would need the wall grammar of Tier 2 applied to the recurrent set rather than to the sector.
- **Open.** The two unidentified pairs, at 1.5537 and 1.7180.
- **Caveat.** `obc0`, and the census is at $N = 10$ with the firing-site scaling checked at $N = 8, 10, 12$. Attractors are monitored-only objects, so everything here is a statement about the measured chain.

Reproduce

```
python -m qca_fragmentation.viz.tower --bc obc0 --N 10
```

writes `fig_tower_obc0` and the two tables. The module is `code/qca_fragmentation/viz/tower.py`; `idle_census(rule, N)` returns the per-rule verdict and `firing_sites(x, N, t)` the raw site set for one state.